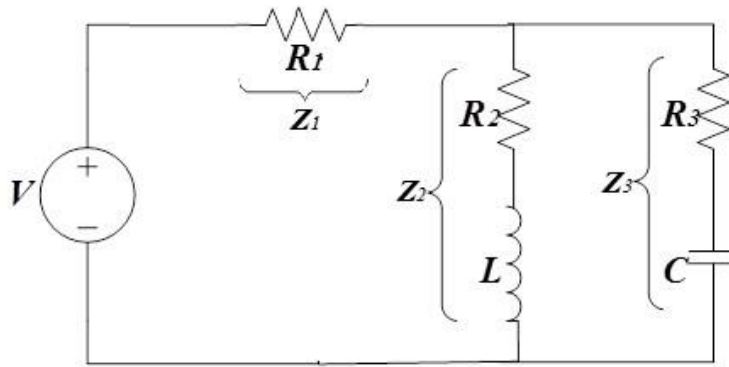


Question 1: The RLC circuit shown below is connected to a voltage supply operating at $f=60\text{ Hz}$.



The parameters of this circuit are given below,

$$V = 120 \cos (120\pi t + 60^\circ),$$

$$R_1 = 15\ \Omega, \quad R_2 = 9\ \Omega, \quad R_3 = 45\ \Omega, \quad L = \frac{1}{12\pi}\text{ H}, \quad C = \frac{1}{15\pi}\text{ F}.$$

Give complete answers to questions about the given circuit up to **4 decimal places**.

- Find the currents passing through R_1 , R_2 , R_3 , C and L .
- Find the complex power at each load (S_{Z1} , S_{Z2} , S_{Z3}), and the power at each element (R_1, R_2, L, R_3, C).

Start your solution here:

